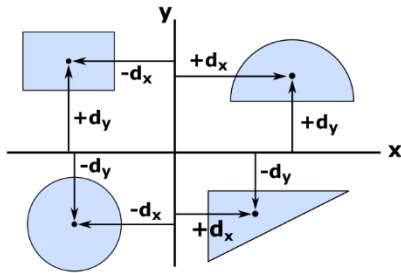


Parallel Axis Theorem (PAT)**for Product of Inertia (POI), I_{xy}**

$$I_{xy} = \bar{I}_{xy} + A d_x d_y$$

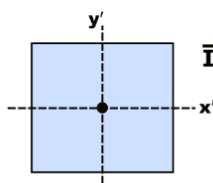
$$I_{xy} = \sum [\bar{I}_{xy} + A d_x d_y]$$

**A = Component Area** **$\bar{I}_{xy} = I_{x'y'}$ = Centroidal POI of Area A****(Note: Some of these $\bar{I}_{xy}$ values are zero or negative. See notes.)** **d_x = Distance from y axis to centroid of A, can be positive or negative!** **d_y = Distance from x axis to centroid of A, can be positive or negative!**

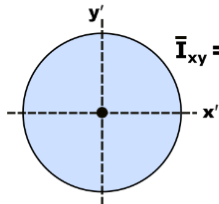
Note: If a sub area A touches the x-y origin, you may use the appropriate I_{xy} "at origin" formula (with the proper + or - sign). This is equivalent to $\bar{I}_{xy} + A d_x d_y$, but easier. See the "at origin" notes.

For shapes symmetric about either the x' or y' axes (centroidal axes)

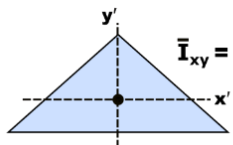
The centroidal POI ($\bar{I}_{xy} = I_{x'y'} = 0$).



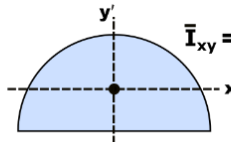
$$\bar{I}_{xy} = I_{x'y'} = 0$$



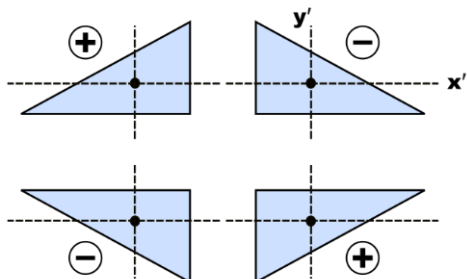
$$\bar{I}_{xy} = I_{x'y'} = 0$$



$$\bar{I}_{xy} = I_{x'y'} = 0$$



$$\bar{I}_{xy} = I_{x'y'} = 0$$

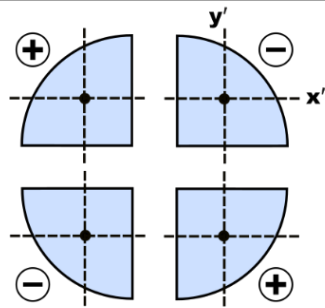


Product of Inertia, $\bar{I}_{xy}$ or $I_{x'y'}$
About Centroidal Axes
Signs vary with orientation:

Right Triangle:

$$\bar{I}_{xy} = I_{x'y'} = \pm \frac{1}{72} b^2 h^2$$

$$= \pm 0.01389 b^2 h^2$$



Product of Inertia, $\bar{I}_{xy}$ or $I_{x'y'}$
About Centroidal Axes
Signs vary with orientation:

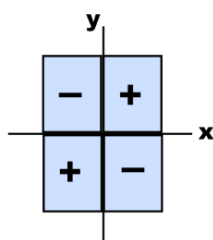
Quarter Circles:

$$\bar{I}_{xy} = I_{x'y'} = \pm 0.01647 R^4$$

Product of Inertia (POI) "at origin" equations

Remember the I_{base} equations for MOI? Product of Inertia (POI) has a similar kind of equation except the sub area A must touch BOTH of the x and y axes at the origin.

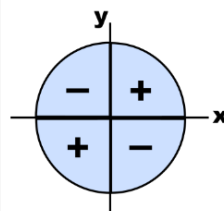
If some of the component areas, A, touch BOTH the x AND y axes, then instead of using the Parallel Axis Theorem (PAT), can use the "at origin" I_{xy} equations for the sub Area, A.
Pay attention to the correct sign!



Rectangle at Origin:

$$I_{xy} = \pm \frac{1}{4} b^2 h^2$$

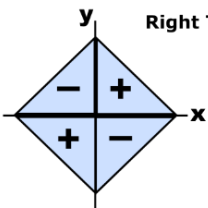
I_{xy} for Shapes at Origin
 Signs Vary by Quadrant



Quarter Circle:

$$I_{xy} = \pm \frac{1}{8} R^4$$

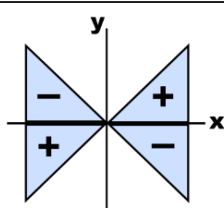
I_{xy} for Shapes at Origin
 Signs Vary by Quadrant



Right Triangle, 90 deg at Origin:

$$I_{xy} = \pm 0.04167 b^2 h^2$$

I_{xy} for Shapes at Origin
 Signs Vary by Quadrant



Right Triangle
Small Vertex at Origin

$$I_{xy} = \pm \frac{1}{8} b^2 h^2$$

I_{xy} for Shapes at Origin
 Signs Vary by Quadrant